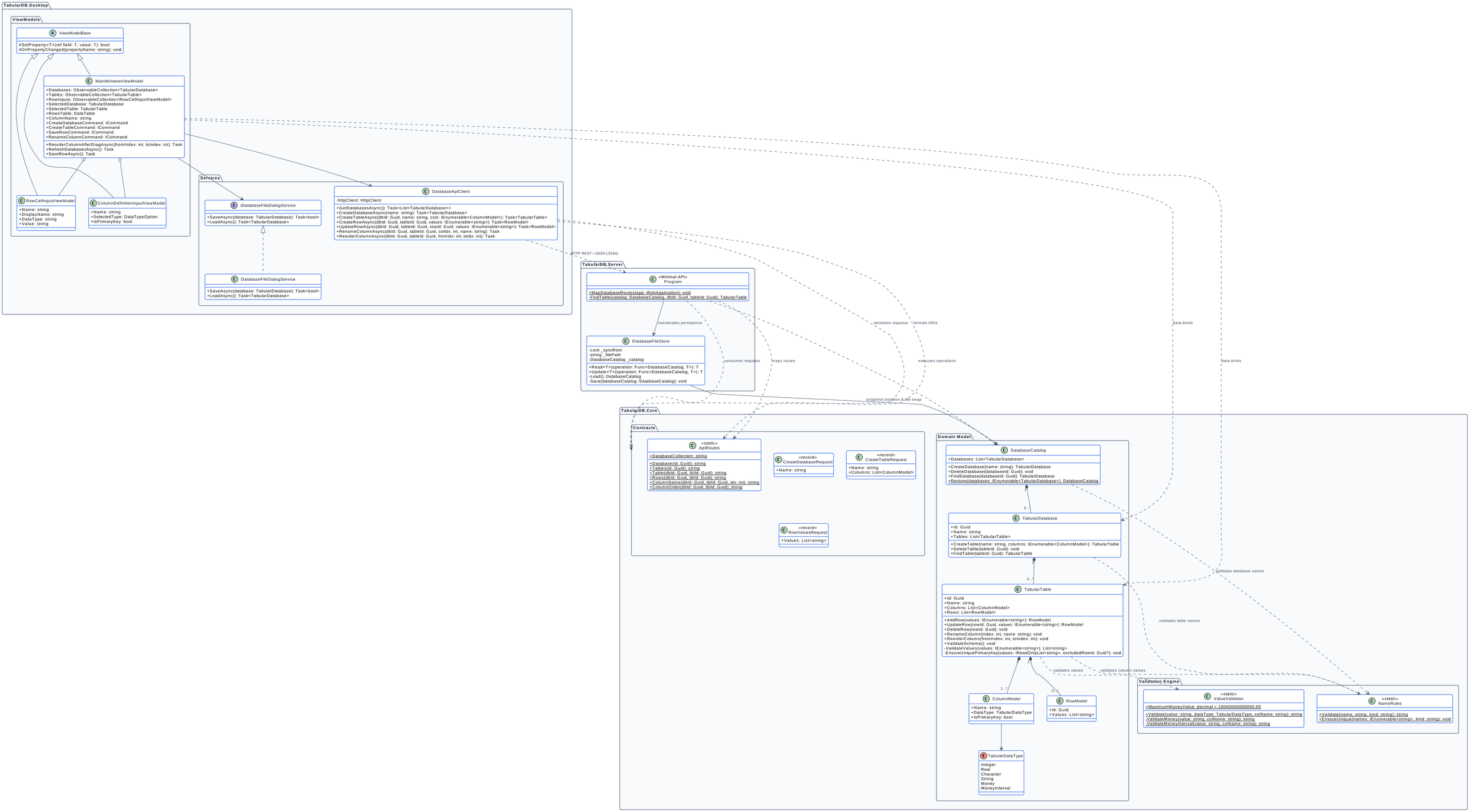
TabularDB.Desktop
ViewModels
ViewModelBase
#SetProperty<T>(ref field: T, value: T): bool
#OnPropertyChanged(propertyName: string): void
MainWindowViewModel
+Databases: ObservableCollection<TabularDatabase>
+Tables: ObservableCollection<TabularTable>
+RowInputs: ObservableCollection<RowCellInputViewModel>
+SelectedDatabase: TabularDatabase
+SelectedTable: TabularTable
+RowsTable: DataTable
+ColumnName: string
+CreateDatabaseCommand: ICommand
+CreateTableCommand: ICommand
+SaveRowCommand: ICommand
+RenameColumnCommand: ICommand
+ReorderColumnAfterDragAsync(fromIndex: int, toIndex: int): Task
+RefreshDatabasesAsync(): Task
+SaveRowAsync(): Task
RowCellInputViewModel
+Name: string
+DisplayName: string
+DataType: string
+Value: string
ColumnDefinitionInputViewModel
+Name: string
+SelectedType: DataTypeOption
+IsPrimaryKey: bool
Services
IDatabaseFileDialogService
+SaveAsync(database: TabularDatabase): Task<bool>
+LoadAsync(): Task<TabularDatabase>
DatabaseFileDialogService
+SaveAsync(database: TabularDatabase): Task<bool>
+LoadAsync(): Task<TabularDatabase>
DatabaseApiClient
-httpClient: HttpClient
+GetDatabasesAsync(): Task<List<TabularDatabase>>
+CreateDatabaseAsync(name: string): Task<TabularDatabase>
+CreateTableAsync(dbId: Guid, name: string, cols: IEnumerable<ColumnModel>): Task<TabularTable>
+CreateRowAsync(dbId: Guid, tableId: Guid, values: IEnumerable<string>): Task<RowModel>
+UpdateRowAsync(dbId: Guid, tableId: Guid, rowId: Guid, values: IEnumerable<string>): Task<RowModel>
+RenameColumnAsync(dbId: Guid, tableId: Guid, colIdx: int, name: string): Task
+ReorderColumnAsync(dbId: Guid, tableId: Guid, fromIdx: int, toIdx: int): Task
HTTP REST / JSON (:5194)
TabularDB.Server
«Minimal API» Program
+MapDatabaseRoutes(app: WebApplication): void
-FindTable(catalog: DatabaseCatalog, dbId: Guid, tableId: Guid): TabularTable
coordinates persistence
DatabaseFileStore
-Lock _syncRoot
-string _filePath
-DatabaseCatalog _catalog
+Read<T>(operation: Func<DatabaseCatalog, T>): T
+Update<T>(operation: Func<DatabaseCatalog, T>): T
-Load(): DatabaseCatalog
-Save(databaseCatalog: DatabaseCatalog): void
serializes requests
formats URIs
consumes requests
maps routes
executes operations
data-binds
data-binds
snapshot isolation & file swap
TabularDB.Core
Contracts
«static» ApiRoutes
+DatabaseCollection: string
+Database(id: Guid): string
+Tables(id: Guid): string
+Table(dbId: Guid, tblId: Guid): string
+Rows(dbId: Guid, tblId: Guid): string
+ColumnName(dbId: Guid, tblId: Guid, idx: int): string
+ColumnOrder(dbId: Guid, tblId: Guid): string
«record» CreateDatabaseRequest
+Name: string
«record» CreateTableRequest
+Name: string
+Columns: List<ColumnModel>
«record» RowValuesRequest
+Values: List<string>
Domain Model
DatabaseCatalog
+Databases: List<TabularDatabase>
+CreateDatabase(name: string): TabularDatabase
+DeleteDatabase(databaseId: Guid): void
+FindDatabase(databaseId: Guid): TabularDatabase
+Restore(databases: IEnumerable<TabularDatabase>): DatabaseCatalog
0..*
TabularDatabase
+Id: Guid
+Name: string
+Tables: List<TabularTable>
+CreateTable(name: string, columns: IEnumerable<ColumnModel>): TabularTable
+DeleteTable(tableId: Guid): void
+FindTable(tableId: Guid): TabularTable
0..*
TabularTable
+Id: Guid
+Name: string
+Columns: List<ColumnModel>
+Rows: List<RowModel>
+AddRow(values: IEnumerable<string>): RowModel
+UpdateRow(rowId: Guid, values: IEnumerable<string>): RowModel
+DeleteRow(rowId: Guid): void
+RenameColumn(index: int, name: string): void
+ReorderColumn(fromIndex: int, toIndex: int): void
+ValidateSchema(): void
-ValidateValues(values: IEnumerable<string>): List<string>
-EnsureUniquePrimaryKey(values: IReadOnlyList<string>, excludedRowId: Guid?): void
validates database names
validates table names
validates values
validates column names
1..*
0..*
ColumnModel
+Name: string
+DataType: TabularDataType
+IsPrimaryKey: bool
RowModel
+Id: Guid
+Values: List<string>
TabularDataType
Integer
Real
Character
String
Money
MoneyInterval
Validation Engine
«static» ValueValidator
+MaximumMoneyValue: decimal = 100000000000000.00
+Validate(value: string, dataType: TabularDataType, colName: string): string
-ValidateMoney(value: string, colName: string): string
-ValidateMoneyInterval(value: string, colName: string): string
«static» NameRules
+Validate(name: string, kind: string): string
+EnsureUnique(names: IEnumerable<string>, kind: string): void